Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**OOA-Based Modular Tooling and Integrated Manufacturing Systems for Large Monolithic Composite Structures in Future Mobility Applications**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		*Mechanical · Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Materials Processing Technologies	*Composite Materials*	*Aerospace Systems*	*Aerospace Composite Structures*
Concept of Proposed Technology	**Core Concept** This technology aims to develop Out-of-Autoclave (OOA) process–based design, structural analysis, and manufacturing technologies for large monolithic composite structures for future mobility applications. The core objective is to establish an integrated manufacturing framework capable of producing large composite structures with uniform curing quality, high structural integrity, and robust process repeatability at full scale. Conventional manufacturing of large composite structures is fundamentally constrained by tooling size limitations, as the fabrication of single-piece large-scale tools is often impractical due to CNC machining capacity, transportation, and handling constraints. As a result, large composite structures are typically segmented, manufactured separately, and subsequently joined, leading to increased structural weight, reduced load transfer efficiency, added manufacturing steps, and lower			

	sustainability. Overcoming these limitations requires a fundamental shift from single-piece tooling to a modular tooling–based manufacturing concept that enables true monolithic fabrication without reliance on oversized tools. The proposed technology addresses the critical technical challenges associated with modular tooling, including the structural stiffness and dimensional stability of tooling under OOA curing conditions, the mechanical and geometric behavior of interfaces between modular tooling segments, and the maintenance of vacuum integrity at tooling joints. These factors directly influence resin flow uniformity, void formation, curing quality, and final structural accuracy in large monolithic composite structures. **Key Functions and Potential Applications** To manage these coupled and scale-dependent phenomena, a digital twin–based simulation framework will be integrated into the modular tooling and manufacturing system [1-2]. The digital twin will model the thermal, mechanical, and vacuum behavior of modular tooling during OOA processing, enabling prediction of tooling deformation, interface leakage risk, and curing-induced geometric variation. By providing simulation-driven insight into tooling design and operational behavior, this approach enables predictive control of the manufacturing process and ensures consistent curing quality and structural reliability as the technology scales to large monolithic structures.
Rationale for Joint R&D Project	**Critical Technical Barriers** The proposed project aims to develop first-in-Korea manufacturing technology for large-scale (≥ 12 m) monolithic composite structures based on OOA processes and modular tooling. While the technical and market need for this technology is clear, the technical risk associated with ≥ 12 m monolithic OOA manufacturing exceeds what can be efficiently mitigated through domestic development alone at the early stages of commercialization, necessitating an international joint R&D approach. At large scale, modular tooling behavior during OOA processing is governed by coupled thermal, mechanical, and vacuum phenomena, including tooling deformation, vacuum integrity at modular interfaces, and curing-induced geometric variation. These effects directly influence curing quality and structural reliability and cannot be fully de-risked through laboratory-scale validation alone. Without access to prior large-scale experience, attempting direct scale-up from TRL 3–4 to full-scale manufacturing would significantly increase the risk of failure, schedule delays, and inefficient use of development resources. **Why Global Collaboration?** International joint R&D enables early risk retirement by leveraging proven experience in modular composite tooling and digital twin–based simulation of tooling and process behavior at scale [3]. Joint verification of tooling structural performance, interface sealing concepts, and simulation-informed operation supports realistic TRL progression from approximately TRL 3 at project initiation to TRL 9 by project completion under industrially relevant conditions. This approach accelerates localization of core manufacturing know-how while ensuring domestic

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	leadership in system integration, manufacturing execution, and commercialization. Securing first-in-Korea capability in the manufacturing of ≥ 12 m monolithic composite structures aligns with the rapid expansion of the global future mobility market, including UAVs, UAM, and eVTOL systems, projected to exceed USD 20 billion by 2030, as well as the domestic market, which is expected to reach approximately KRW 13 trillion by 2040. Early technology securing between 2026 and 2030 positions the project to support industrial leadership, supply chain resilience, and sustainable participation in global future mobility markets.
Key Words	1. Modular Composite Tooling 2. Large-Scale Monolithic Composite Structures 3. Out-of-Autoclave (OOA) Manufacturing 4. Digital Twin–Based Tooling Operation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives	**R&D Objectives** This project aims to integrate and develop manufacturing technologies for large monolithic composite structures based on OOA processes, with the goal of achieving significant structural weight reduction and productivity enhancement for future mobility platforms. The development scope encompasses monolithic composite manufacturing technologies, modular composite tooling, a mobile ground operation platform, and a digital twin–based process and tooling operation framework, all designed and validated as a single, unified, and interoperable manufacturing system. **Detailed Objectives** ● Secure manufacturing technologies for large monolithic composite structures with a length of 12 meters or more, applicable to future mobility platforms such as UAVs, UAM, and eVTOL systems. ● Develop modular tooling technologies dedicated to the fabrication of large monolithic composite components, incorporating structural stability, interface integrity, and dimensional control. ● Develop a digital twin–based simulation and operational framework to predict and manage tooling deformation, vacuum integrity, and curing-induced geometric variation during OOA processing. ● Develop a dedicated mobile manufacturing and ground operation platform to support modular tooling–based fabrication, handling, and curing operations. ● Establish large-scale OOA curing infrastructure integrated with the mobile platform and digital twin framework to ensure uniform curing quality, dimensional accuracy, and process repeatability at full scale.			

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R&D Target Specifications

No	KPI	Unit	Target Value	Measurement and Verification Method
1	Length of monolithic composite structure manufactured	m	≥ 12	Verification through fabrication and physical measurement of the completed full-scale monolithic structure
2	Reduction rate in the number of components	%	≥ 30	Comparison of part count and joint elements with conventional segmented manufacturing methods
3	Geometric deviation (warpage) of a 12 m-class structure	deg	≤ 1.0	Measurement of geometric deviation relative to the design using a 3D scanning or laser tracker system
4	Thickness tolerance accuracy of a 12 m-class structure	%	± 5	Ultrasonic thickness measurement of major structural regions compared with design specifications
5	Vacuum sealing performance at modular tooling interfaces	inchHg	≥ 22	Evaluation of vacuum retention performance at tooling interfaces during OOA curing (e.g., vacuum drop test)
6	Digital twin prediction accuracy for tooling deformation and part geometry	%	≥ 90	Comparison between digital twin simulation results and measured tooling deformation and part geometry obtained from full-scale manufacturing

R&D Plan

Year 1: Core Technology Foundation and Digital Twin Framework Development

- Selection of composite materials suitable for large-scale OOA manufacturing and evaluation of fundamental material properties through standardized testing.
- Establishment of large-scale OOA processing infrastructure, including heating, vacuum, and process monitoring systems.
- Development of a digital twin framework for modular tooling and OOA processing, focusing on thermal behavior, tooling deformation, vacuum integrity, and curing-induced geometric variation.
- Conceptual design of a dedicated mobile manufacturing and curing platform for large composite structures.

Year 2: Structural Design, Modular Tooling Development, and Process Integration

- Structural design and analysis of large monolithic composite structures (≥ 12 m class) for future mobility applications.
- Design and material selection for modular tooling, including structural and thermal analysis considering thermal expansion and interface behavior.
- Fabrication of modular tooling segments and integration with the mobile manufacturing platform.
- Refinement of the digital twin through correlation with tooling fabrication data and intermediate-scale manufacturing trials.
- Development of integrated OOA process parameters for modular tooling–based monolithic manufacturing.

Year 3: Full-Scale Manufacturing, Validation, and Performance Verification

- Full-scale manufacturing of a ≥ 12 m monolithic composite structure using modular tooling, mobile platform, and large-scale OOA curing infrastructure.
- Validation of digital twin prediction accuracy by comparing simulated tooling deformation,

	vacuum behavior, and part geometry with measured results from full-scale manufacturing. ● Process optimization based on simulation–experiment correlation to ensure uniform curing quality and dimensional accuracy. ● Conducting static structural tests to verify load-bearing capability and structural integrity. ● Conducting dynamic structural tests to validate durability, vibration characteristics, and operational reliability under representative operating conditions.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The proposed project adopts an international collaborative research approach with the Composites Manufacturing and Simulation Center (CMSC) at Purdue University, which possesses leading expertise in modular composite tooling, digital twin–based manufacturing simulation, and process–structure integration. CMSC will lead tooling-level technology development and digital twin framework development for large-scale OOA manufacturing [1-3]. Specifically, CMSC will be responsible for simulation-driven design and analysis of modular tooling, including prediction of thermal expansion, tooling deformation, vacuum integrity at modular interfaces, and curing-induced geometric variation during OOA processing. CMSC will support tooling-level fabrication and demonstration activities to validate simulation predictions and retire key technical risks prior to full-scale manufacturing. By assigning CMSC responsibility for digital twin development, simulation-informed tooling design, and early-stage risk retirement, while retaining domestic leadership in system integration and OOA process development, the project minimizes development risk, accelerates technology maturation, and enables effective technology transfer and localization of advanced composite manufacturing capabilities.
Expected R&R of Korean SME Partner	Korean SME will be responsible for full-scale fabrication, inspection, and demonstration of large monolithic composite structures using the modular tooling, mobile manufacturing platform, and large-scale OOA curing infrastructure developed through the project. Full-scale manufacturing activities will be conducted under industrially relevant conditions to validate process repeatability, curing quality uniformity, and structural reliability. In addition, Korean SME will perform comprehensive full-scale part inspection and deviation analysis, including geometric accuracy, warpage, thickness distribution, and interface quality assessment using 3D scanning, laser tracking, and non-destructive evaluation methods. Measured results will be used to assess manufacturing accuracy and to provide feedback for correlation with digital twin predictions. Korean SME will commit the engineering personnel, manufacturing facilities, testing infrastructure, and quality inspection resources required for large-scale composite fabrication and validation. Korean SME will also lead project management and commercialization-oriented activities, including schedule control, risk management, coordination among project partners, and compliance with project milestones, ensuring a smooth transition from technology development to industrial deployment.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Technical Impact** ● Securing Core Technology Assets through Establishment of a Large-Scale Composite Manufacturing System Through this project, an integrated manufacturing framework encompassing design, structural analysis, fabrication, and process management for large composite structures will be established. This end-to-end system capability is expected to be accumulated as a company-specific core technology asset, strengthening long-term technological competitiveness in large composite structure manufacturing. ● Localization of Modular Tooling Technology through International Collaboration By leveraging international collaboration with leading overseas research institutions, advanced modular tooling design, analysis, and manufacturing technologies will be transferred and localized to suit domestic industrial environments. This will enhance technological self-reliance in large composite structure manufacturing while reducing long-term dependence on foreign tooling and process technologies. ● Expandability of Large Composite Manufacturing Technology to Other Industries The developed technology is not limited to UAVs or future air mobility platforms, but is broadly applicable to industries requiring large composite structures, including unmanned surface vessels, unmanned underwater vehicles, next-generation hybrid air–sea platforms, and defense applications. As a result, the technology is expected to demonstrate high versatility and generate significant industrial ripple effects across multiple sectors.
	Expected Economic Impact ● Early Market Entry and Leadership in the Domestic Future Mobility Structural Manufacturing Market Based on the developed technology, the company aims to secure over 50% market share in the domestic future mobility (UAM, UAV, etc.) structural manufacturing market after 2030. This is expected to establish a leading domestic supplier position in large composite structure manufacturing, enabling high value-added industrial growth and a stable revenue base. ● Global Market Expansion and Overseas Revenue Generation Based on DOA–POA Capabilities In overseas markets, a design–manufacturing–certification–linked cooperation model utilizing DOA (Design Organization Approval) and POA (Production Organization Approval) capabilities will be pursued. Through direct participation in the global aviation and future mobility supply chain, the company targets securing more than 5% of the global future mobility structural manufacturing market after 2030, leading to sustainable overseas revenue generation. ● Regional Employment Creation and Industrial Ecosystem Revitalization During the technology development and advancement phases, professionals will be

	recruited and trained locally. This will expand the talent pool in large composite structure design, analysis, and manufacturing, while contributing to regional industrial ecosystem revitalization and sustainable workforce development.

6. References

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(2) Barocio, E., Yu, W., Pipes, R. B. (2019), “Simulation-driven design and manufacturing of composite structures.” Mechanics of Advanced Materials and Structures.

(3) U.S. Department of Energy (DOE) NEPA CX Determination (2024), “Additive Manufacturing of Modular Tools with Integrated Heating for Large-Scale Wind Blade Manufacturing.” Purdue University.